PREDICTIVE ANALYTICS
PROBLEM SET 3
MULTIPLE LINEAR REGRESSION

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2.Problem to demonstrate the role of qualitative (nominal) predictors in addition to quantitative predictors in multiple linear regression

Attach "Credits" data from R.

```
library(ISLR)
attach(Credit)
```

Regress "balance" on

(a) "gender" only.

```
m1<-lm(Balance~Gender)
```

(b) "gender" and "ethnicity" .

```
m2<-lm(Balance~Gender+Ethnicity)
```

(c) "gender", "ethnicity", "income".

```
m3<-lm(Balance~Gender+Ethnicity+Income)
```

(d) Output all the regressions in (a)-(c) in a single table using stargazer.

```
library(stargazer)

## Warning: package 'stargazer' was built under R version 4.5.2

##
## Please cite as:

##  Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics
Tables.

##  R package version 5.2.3. https://CRAN.R-project.org/package=stargazer

stargazer(m1,m2,m3,type='text',out='f1.txt')

##
## ===============================================================================
##                                          Dependent variable:
##                      -------------------------------------------------------------
##                                                 Balance
##                             (1)                 (2)                   (3)
## -------------------------------------------------------------------------------
## GenderFemale              19.733              20.038                24.340
##                          (46.051)            (46.178)              (40.963)
##
## EthnicityAsian                                -19.371                1.637
##                                              (65.107)              (57.787)
```

```
##
## EthnicityCaucasian                          -12.653                6.447
##                                            (56.740)              (50.363)
##
## Income                                                           6.054***
##                                                                   (0.582)
##
## Constant                 509.803***          520.880***            230.029***
##                           (33.128)            (51.901)              (53.857)
##
## ---------------------------------------------------------------------------------
## Observations                400                 400                   400
## R2                        0.0005               0.001                 0.216
## Adjusted R2               -0.002              -0.007                 0.208
## Residual Std. Error 460.230 (df = 398)  461.337 (df = 396)    409.218 (df = 395)
## F Statistic         0.184 (df = 1; 398) 0.092 (df = 3; 396) 27.161*** (df = 4; 395)
## =================================================================================
## Note:                                                  *p<0.1; **p<0.05; ***p<0.01
```

Comment on the significant coefficients in each of the models.

(e) Explain how gender affects "balance" in each of the models (a)- (c).

(f) Compare the average credit card balance of a male African with a male Caucasian on the basis of model (b).

```
#model_b_estimated_balance=520.880+20.038*GenderFemale-19.371*EthnicityAsian-
12.653*EthnicityCaucasian
```

Comment:

Here, GenderFemale(=1,if female,=0,if male),

EthnicityAsian(=1, if Asian,=0, if not),

EthnicityCaucasian(=1, if Caucasian, =0, if not),

for Ethnicity, African is the baseline category.

Thus for Male African, we have to put on the above model-

GenderFemale=0,EthnicityAsian=0,EthnicityCaucasian=0,leading to the average balance-520.880$.\par
Thus for Male Caucasian, we have to put on the above model-\par
GenderFemale=0,EthnicityAsian=0,EthnicityCaucasian=1,leading to the average balance-(520.880-12.653)$

Thus we say, on the basis of model in (b), the average credit balance of a customer decreases by 12.653$ when the ethnicity changes from African to Caucasian.

(g) Compare the average credit card balance of a male African with a male Caucasian when each earns 100,000 dollars. For comparison, use the model in (c).

```
#model_c_estimated_balance=230.029+24.340*GenderFemale+1.637*EthnicityAsian+6.447*Ethn
icityCaucasian+6.054*Income
```

Comment:

Here, GenderFemale(=1,if female,=0,if male),

EthnicityAsian(=1, if Asian,=0, if not),

EthnicityCaucasian(=1, if Caucasian, =0, if not),

for Ethnicity, African is the baseline category,

Income:Income in $1000's.\par Thus for Male African, we have to put on the above model-\par GenderFemale=0,EthnicityAsian=0,EthnicityCaucasian=0,Income=100,leading to the average balance-\par (230.029+6.054*100)$.

Thus for Male Caucasian, we have to put on the above model-

GenderFemale=0,EthnicityAsian=0,EthnicityCaucasian=1,Income=100,leading to the average balance-

(230.029+6.447+6.054*100)$.\par Thus we say, on the basis of model in (c), the average credit balance of a customer increases by (6.447)$ when the ethnicity changes from African to Caucasian given both earn 100,000$.

(h) Compare and comment on the answers in (f) and (g).

Inclusion of the Income predictor changes the coefficient of Ethnicity predictors from negative to positive leading to drastic change in conclusion.

(i) Based on the model in (c), predict the credit card balance of a female Asian whose income is 2000,000 dollars.

```
#model_c_estimated_balance=230.029+24.340*GenderFemale+1.637*EthnicityAsian+6.447*Ethn
icityCaucasian+6.054*Income
```

Comment:

Here, GenderFemale(=1,if female,=0,if male),

EthnicityAsian(=1, if Asian,=0, if not),

EthnicityCaucasian(=1, if Caucasian, =0, if not),

for Ethnicity, African is the baseline category,

Income:Income in $1000's.\par Thus for Female Asian, we have to put on the above model-\par GenderFemale=1,EthnicityAsian=1,EthnicityCaucasian=0,Income=2000,leading to the average balance-\par (230.029+24.340+1.6376.054*2000)$=12364.01$.

```
230.029+24.340+1.637+(6.054*2000)

## [1] 12364.01
```

(j) Check the goodness of fit of the different models in (a)-(c) in terms of AIC,BIC and adjusted R2. Which model would you prefer?

Comment: In terms of Adjusted R^2, model c has a positive and much greater value leading to its preference over the others.

4.Problem to demonstrate the impact of ignoring interaction term in multiple linear regression

Consider a simulation setting where the data is generated as follows:

```
n=100
beta_0=c(-2.5,-2.5)
beta_1=c(1.2,1.2)
beta_2=c(2.3,2.3)
beta_3=c(0.001,3.1)
```

Step 1: Generate x1i from Normal(0,1) distribution, i = 1,2,..,n

```r
set.seed(123)
x1i=rnorm(n)
```

Step 2: Generate x2i from Bernoulli (0.3) distribution, i = 1,2,..,n

```r
set.seed(123)
x2i=rbinom(n,size=1,prob=0.3)
```

Step 3: Generate ei from Normal(0,1) and hence generate the response yi =beta_0 +beta_1x1i +beta_2x2i +beta_3(x1i × x2i)+ei, i = 1,2,,,n.

```r
set.seed(123)
ei=rnorm(n)
yi1=beta_0[1]+(beta_1[1]*x1i)+(beta_2[1]*x2i)+(beta_3[1]*(x1i*x2i))+ei;yi1

##   [1] -3.73304642 -0.70662065  0.92915829 -0.04481103  0.08456231  1.27314297
##   [7] -1.48598435 -2.98439978 -4.01107627 -3.48045633  2.49420404 -1.70840958
##  [13] -1.61830281 -2.25649802 -3.72285050  3.73299581 -1.40472895 -6.82655774
##  [19] -0.95701702 -1.24061389 -2.55027998 -2.97954481 -4.75720979 -1.80428960
##  [25] -3.87508639 -3.91241198 -0.65686850 -2.16257914 -5.00390126  0.25839283
##  [31]  0.73864775 -0.84945233 -0.53072355  1.73277181 -0.69252162 -0.98499144
##  [37]  1.01917276 -2.63620576 -3.17311786 -3.33703620 -4.02835535 -2.95741801
##  [43] -5.28387197  2.27170312  0.15751640 -4.97083888 -3.38634664 -3.52664178
##  [49] -0.78407674 -0.38349532 -1.94269927 -2.56280286 -0.29435788  0.51092502
##  [55] -2.99669617  0.83623533 -5.90725617  1.08673486  0.07260319 -2.02492855
##  [61] -1.66479314 -3.60511160 -3.23305624 -4.74086584 -2.55901249 -1.83223699
##  [67]  0.78650972 -0.08333770  1.82991070  2.01018631 -1.28075960 -7.58017153
##  [73]  2.01363049 -4.06024168 -4.01361896 -0.24374299 -3.12650062 -5.18557897
##  [79] -2.10113234 -2.80556100 -2.48731879 -1.65238312 -3.31545207  1.21827278
##  [85] -2.98507044 -1.77007968  2.21414267  0.75783446 -0.91737542  0.02737676
##  [91] -0.31429152 -1.29352669 -1.97479018 -3.88139337  0.49343539 -3.82057109
##  [97]  4.61431992  0.87174338 -3.01854079 -4.75812598

yi2=beta_0[2]+(beta_1[2]*x1i)+(beta_2[2]*x2i)+(beta_3[2]*(x1i*x2i))+ei;yi2

##   [1] -3.73304642 -1.41994069  0.92915829  0.17369447  0.48522500  1.27314297
##   [7] -1.48598435 -6.90482454 -4.01107627 -3.48045633  6.28763353 -1.70840958
##  [13] -1.61830281 -2.25649802 -3.72285050  9.27063963 -1.40472895 -6.82655774
##  [19] -0.95701702 -2.70579446 -5.85946564 -2.97954481 -4.75720979 -4.06312352
##  [25] -3.87508639 -9.13947455 -0.65686850 -2.16257914 -5.00390126  0.25839283
##  [31]  2.06026037 -1.76387886 -0.53072355  4.45410748 -0.69252162 -0.98499144
##  [37]  2.73576356 -2.63620576 -3.17311786 -3.33703620 -4.02835535 -2.95741801
##  [43] -5.28387197  2.27170312  0.15751640 -4.97083888 -3.38634664 -3.52664178
##  [49] -0.78407674 -0.64185605 -1.94269927 -2.56280286 -0.42721342  0.51092502
##  [55] -2.99669617  0.83623533 -5.90725617  2.89845287  0.45642749 -2.02492855
##  [61] -1.66479314 -3.60511160 -3.23305624 -4.74086584 -5.88049350 -1.83223699
##  [67]  2.17551183  0.08092240  4.68801758  2.01018631 -2.80246518 -7.58017153
##  [73]  5.13041418 -4.06024168 -4.01361896 -0.24374299 -3.12650062 -5.18557897
##  [79] -2.10113234 -2.80556100 -2.48731879 -1.65238312 -3.31545207  3.21519571
##  [85] -2.98507044 -1.77007968  5.61324677  2.10646190 -1.92743740  0.02737676
##  [91] -0.31429152 -1.29352669 -1.97479018 -3.88139337  0.49343539 -3.82057109
##  [97] 11.39286486  0.87174338 -3.01854079 -4.75812598
```

Step 4: Run two separate multiple linear regressions (i) using the model in Step 3 and (ii) using the model in Step 3 without the interaction term.

```r
fit1=lm(yi1~x1i+x2i+x1i*x2i)
fit2=lm(yi1~x1i+x2i)
library(stargazer)
stargazer(fit1,fit2,type='text',out='f2.txt')
```

```
##
##
==========================================================================================
================================
##                                                                  Dependent variable:
##                      ---------------------------------------------------------------
----------------------------------
##                                                                            yi1
##                                                         (1)
(2)
## ------------------------------------------------------------------------------------
----------------------------------
## x1i                                                   2.200***
2.200***
##                                                        (0.000)
(0.00005)
##
## x2i                                                   2.300***
2.300***
##                                                        (0.000)
(0.0001)
##
## x1i:x2i                                               0.001***
##                                                        (0.000)
##
## Constant                                              -2.500***
-2.500***
##                                                        (0.000)
(0.00005)
##
## ------------------------------------------------------------------------------------
----------------------------------
## Observations                                            100
100
## R2                                                     1.000
1.000
## Adjusted R2                                            1.000
1.000
## Residual Std. Error                           0.000 (df = 96)
0.0004 (df = 97)
## F Statistic         481,644,987,064,531,068,640,882,062,660,026.000*** (df = 3; 96)
1,588,256,965.000*** (df = 2; 97)
##
==========================================================================================
================================
## Note:
*p<0.1; **p<0.05; ***p<0.01
```

Repeat Steps 1-4 , R = 1000 times. At each simulation compute the MSE for the correct model (i.e. model with the interaction term) and the naive model (i.e. the model without the interaction term). Finally find the average MSE's for each model. From the output, demonstrate the impact of ignoring the interaction term.

Carry out the analysis for n = 100 and the following parametric configurations:
(beta_0,beta_1,beta_2,beta_3) = (−2.5,1.2,2.3,0.001) , (-2.5, 1.2, 2.3, 3.1). Set seed as 123.

```
set.seed(123)

simulation <- function(beta_0, beta_1, beta_2, beta_3, n=100, R=1000){
```

```r
  mse_correct <- numeric(R)
  mse_naive   <- numeric(R)

  for(r in 1:R){

    x1 <- rnorm(n,0,1)
    x2 <- rbinom(n,1,0.3)
    e <- rnorm(n,0,1)

    y <- beta_0 + beta_1*x1 + beta_2*x2 + beta_3*(x1*x2) + e

    # Correct model
    fit1 <- lm(y ~ x1+x2+(x1*x2))
    mse_correct[r] <- mean(residuals(fit1)^2)

    # Naive model (no interaction)
    fit2 <- lm(y ~ x1 + x2)
    mse_naive[r] <- mean(residuals(fit2)^2)
  }

  c(mean(mse_correct), mean(mse_naive))
}

simulation(-2.5,1.2,2.3,0.001)

## [1] 0.9631944 0.9739083

simulation(-2.5,1.2,2.3,3.1)

## [1] 0.9577982 2.8633349
```

Comment: 1.The MSE of Correct Model (with interaction term) is much less than MSE of Naive Model (without interaction term).

2.When the coefficient of interaction term rises from 0.001 to 3.1, the increment in MSE from Correct to Naive Model rises drastically.